Assignment 1

1) $f(x) = \frac{e^{2x} - 1}{x}$ at $a = -0{,}5$

$$f(-0{,}5) = \frac{e^{-1} - 1}{-0{,}5} \approx \frac{0{,}36 - 1}{-0{,}5} = \frac{-0{,}64}{-0{,}5} \approx 1{,}28$$

$$f'(x) = \frac{2e^{2x} \cdot x - e^{2x} + 1}{x^2}$$

$$f'(-0{,}5) = \frac{2 \cdot e^{-1} \cdot (-0{,}5) - e^{-1} + 1}{0{,}25} = \frac{-0{,}36 - 0{,}36 + 1}{0{,}25}$$

$$\approx \frac{-0{,}72 + 1}{0{,}25} \approx \frac{0{,}28}{0{,}25} \approx 1{,}12$$

$$f''(x) = \frac{x \cdot 4e^{2x} \cdot x^2 - 2x(2xe^{2x} - e^{2x} + 1)}{x^3} \approx -0{,}8$$

$u = x$ $\quad uv = u'v + v'u$

$v = 2e^{2x}$ $\quad (2xe^{2x})' = 2e^{2x} + x \cdot 4e^{2x}$

$$(2e^{2x} \cdot x - e^{2x} + 1)' = \cancel{2e^{2x}} + x \cdot 4e^{2x} - \cancel{2e^{2x}} + 0$$

$$= x \cdot 4e^{2x}$$

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Taylor series: $1{,}28 + \frac{1{,}12(x+0{,}5)}{1!} - \frac{0{,}8(x+0{,}5)}{2!}$

2) $f(x) = \sin 2x \qquad a = 0{,}5\pi$

$f(0{,}5\pi) = \sin \pi = 0$

$f'(x) = 2\cos 2x$

$f'(0{,}5\pi) = 2\cos \pi = -2$

$f''(x) = \cancel{-4\cos} -4\sin 2x$

$f''(0{,}5\pi) = -4\sin \pi = 0$

Taylor series: $0 - \frac{2(x-0{,}5\pi)}{1!} + 0$

$= -2(x-0{,}5\pi)$